

Nini. Modeling QC Feedback

Asset	Nini (character)
Reviewed file	D:\RIGS\Nini\AUR_Nini_Modeling_v0001.fbx (FBX, 60 meshes, 61,257 vertices)
Reference	Modeling QC Checklist (Topology/Model design checks, Tech checks)
Reviewed by	Rigging department
Date	16 September 2026
Status	NOT APPROVED. Fix the items in section 2 and resubmit. Items in section 3 are recommended.

1. Summary

The model is clean in most technical aspects: perfectly symmetrical body, no construction history, no non-manifold or lamina geometry, no duplicate names, no illegal nodes, one shape per transform, neutral face expression, legs parallel and facing +Z, and both limb chains pass the single-axis plane test with a small deviation (section 4). Five things block rigging: unmerged vertices along the mirror seam (they produce n-gons and duplicate vertices on the body, hood and pants, and break the lip sealing loop), unfrozen transforms on several groups and meshes, inconsistent naming, locked vertex normals on every mesh, and the delivery format (FBX instead of a Maya scene).



Front view (+Z facing camera).



Side view. Character faces +Z.

2. Must fix (blockers)

2.1 Unmerged center-seam vertices, n-gons and duplicate vertices

The mirror seam was not merged at several points. Each unmerged pair leaves two vertices at the same position, a valence-2 vertex and two n-gons around it. Merge the seam (Edit Mesh > Merge with a small threshold, or Mesh > Mirror with merge) and re-check the loops.

Mesh	Location (world cm)	Issue
c_body_GEO	(0, 110.3, 10.5) and (0, 112.0, 11.0)	Center of the lips. 2 duplicate pairs, 4 n-gons. The lip sealing loop is cut at the center.
c_body_GEO	(0, 113.9, -6.7)	Back of the neck. 1 duplicate pair, 2 n-gons.
c_body_GEO	(0, 105.1, 2.3), (0, 107.3, 3.6), (0, 108.0, 6.2)	Inside the mouth cavity. 2 duplicate pairs, 6 n-gons.
c_hood_GEO1	(0, 122.8, -11.9)	Back center of the hood. 2 n-gons.
c_pants_GEO	(0, 64.6, -11.3)	Back center of the pants. 2 n-gons, 1 duplicate pair.

Mesh	Location (world cm)	Issue
c_button_GEO	2 duplicate vertices	Merge.

Totals: 16 n-gons and 8 duplicate vertex pairs. The checklist requires zero n-gons on deforming meshes and no two vertices in the same position.

2.2 Transforms not frozen, pivots not at the origin

Every transform must have translate 0, rotate 0, scale 1 and its pivot at the world origin. These nodes fail:

Node	Translate	Rotate	Scale
nini_GRP	0, 0, 2	0	1
c_hair_GEO (group)	0, 0, -2	0	1
c_eyes_GRP	4.88, 119.78, 4.84	-5.03, 13.13, 0.34	2.386
c_eyelashes_GEO	-1.82, -49.78, -6.91	5.25, -13.05, -1.52	0.419
teeth_top (group)	0, 0.23, -0.50	0	1.163, 1.102, 1.102
teeth_low (group)	0, -1.33, -0.77	175.63, 180.34, -0.45	1.163, 1.317, 1.102
c_placeholder_bootBase	31.85, 0.07, 0	0	1, 1.125, 1
R_specular_GEO	-0.30, 0.01, -0.07	0	1
c_furBalls_GEO	0, -0.87, 1.97	0	1

The nini_GRP offset of +2 in Z is cancelled by -2 on c_hair_GEO, so the hair is the only thing that sits where it should relative to the file. Freeze everything and put all pivots at (0, 0, 0). Most mesh pivots are currently at the mesh center (for example c_body_GEO at 0, 66.26, 0.49).

2.3 Naming convention

Node	Problem	Suggested name
c_hood_GEO1	Numeric suffix after _GEO.	c_hood_GEO
c_placeholder_bootBase	No _GEO suffix. Confirm this is final geometry, not a placeholder.	c_bootBase_GEO
c_hair_GEO	It is a group, not a mesh.	c_hair_GRP
teeth_top, teeth_low	Groups without side prefix and _GRP suffix.	c_teethTop_GRP, c_teethLow_GRP
C_tongue_GEO	Upper-case C_ while the rest uses c_.	c_tongue_GEO
L_/R_ eyes and teeth vs l_/r_ eyebrows	Mixed case for side prefixes.	Pick one case (l_/r_ or L_/R_) for the whole asset.

2.4 Locked vertex normals on all 60 meshes

Every mesh has locked per-vertex normals. Locked normals do not update when the mesh deforms, so the shading breaks as soon as the rig moves. This is a side effect of the FBX export. Unlock them in Maya (Mesh Display > Unlock Normals) or deliver the Maya scene directly.

2.5 Delivery format

Only an FBX was delivered. Deliver a Maya ASCII (.ma) scene. The FBX import also created 6 groupId nodes and 3 shading groups per pupil (per-face material assignment on L_pupil_GEO and R_pupil_GEO). Assign one material per mesh unless the per-face split is required by lookdev.

3. Should fix (recommended)

3.1 Poles on deformation areas

Poles (vertices with 3 or 5 edges) on a fold line produce pinching when the joint bends. The knees are clean. These areas need attention:

Area	Finding	Recommendation
Elbows (both sides)	4 poles within 7 cm of the elbow center. Two valence-5 vertices sit on the elbow loop itself, at (22.4, 84.3, -4.9) and (25.8, 87.6, -5.4).	Move the poles above or below the fold. The fold loop must be a straight quad loop.
Shoulders (both sides)	9 poles within 7 cm of the arm root.	Acceptable if they stay off the armpit crease. Check after freezing.
Mouth corners	Valence-5 vertex at each commissure (4.6, 110.6, 7.8) plus a cluster of poles on the philtrum.	Keep poles away from the commissure. The corner needs a clean loop to open the jaw.
Wrists	78 poles near the hand. The hand is a mitten shape with dense topology.	Fine for a mitten hand. No action if the hand stays inside the glove.

3.2 Lips and eyelids

The expression is neutral and the face is symmetrical. The lip sealing loop is currently cut at the center by the unmerged vertices (item 2.1), so the upper/lower vertex match could not be validated. After merging the seam, confirm that the same circular loop runs through the lip contact line and through the eyelid contact line, and that upper and lower lips and eyelids have the same vertex count. This will be re-checked on resubmission with a single jaw joint skin test.

3.3 Mesh count and GPU evaluation

57 of the 60 meshes are under 2,000 vertices, including 13 separate meshes per dental arch. Deformers on meshes below 2,000 vertices run on the CPU. Combine the upper teeth into one mesh and the lower teeth into one mesh (keep gums separate if they need their own deformation). Consider combining the 5 hair meshes if they share the same deformation.

3.4 Ground contact

Lowest point of the boot base is at $Y = 0.054$ and the body at $Y = 0.386$. The character is 0.05 cm above the ground. Snap the sole to $Y = 0$.

3.5 Knee pre-bend

The knee sits 5.0 cm in front of the hip-ankle line, an interior angle of 157 degrees. The direction is correct (+Z) and a pre-bend is desirable for IK, but this amount is on the high side. A 2 to 3 cm offset is enough. Keep it if the design calls for it.

3.6 Limb plane alignment

Both limbs pass the single-axis plane test (section 4) with a small deviation. Two optional corrections would make the chains exact: move the knee loop 0.5 cm toward the center line so hip, knee and ankle share the same X (removes the 2.2 degree residual and the 1.0 cm ankle drift); rotate the forearm around the upper-arm axis so the elbow points straight back (-Z) with no downward tilt, keeping the A-pose angle (removes the 4 degree residual and the 1.6 cm wrist drift). The knee pre-bend of 22.5 degrees can also be reduced to 5 to 10 degrees, see 3.5.

4. Limb plane check (single rotation axis)

Method: a 3-joint chain was placed on each limb (hip, knee, ankle and shoulder, elbow, wrist) at the centroid of the body mesh cross-sections. The chain was oriented with Orient Joint (primary X down the chain, secondary axis toward the expected bending plane normal: world X for the leg, world -Z for the arm). A limb is in one plane when the middle joint carries rotation on one axis only. The root can have any orientation. The test then zeroes the two minor axes of the middle joint and measures how far the end guide moves from its landmark. Pass condition: with

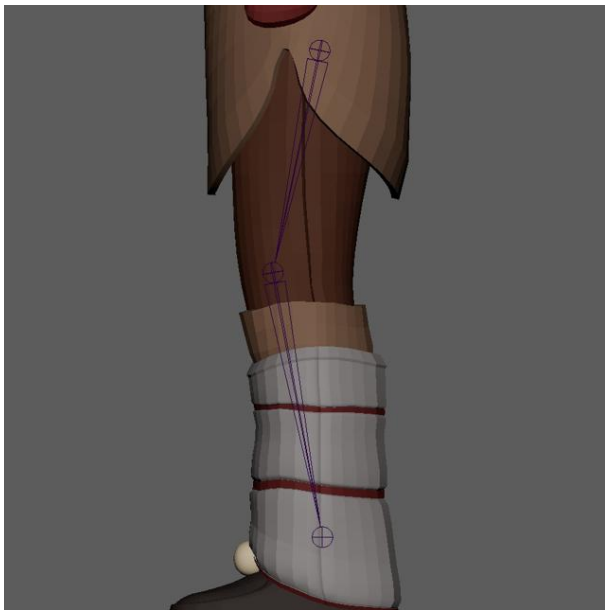
the minor axes zeroed, the end guide must stay approximately in place, inside the ankle or wrist volume and close to its center.

Chain	Middle joint orient (X, Y, Z) after Orient Joint	Orient with the other axes at 0	Residual removed	End guide drift after zeroing	Pass
Leg (L and R)	knee: -0.45, 22.48, -2.20	knee: 0, 22.48, 0 (Y only)	0.45 deg X, 2.20 deg Z	ankle moves 1.0 cm in X (ankle radius about 2.4 cm)	YES, small deviation
Arm (L and R)	elbow: 0.95, -3.99, -17.60	elbow: 0, 0, -17.60 (Z only)	0.95 deg X, 3.99 deg Y	wrist moves 1.6 cm (1.15 X, 1.1 Y; wrist radius about 2.9 cm)	YES, small deviation

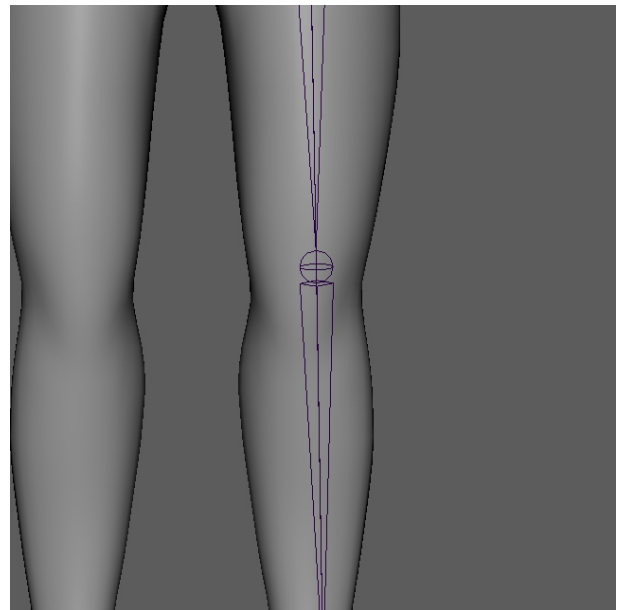
Leg: the knee center sits 0.47 cm outward (X) from the hip-ankle line, which shows up as 2.2 degrees on the Z axis of the knee. With that axis zeroed the ankle guide lands 1.0 cm off center, still inside the ankle. Accepted. The knee points +Z (5.4 degrees off). Optional correction in 3.6.

Arm: the elbow bends backward but the bending plane is tilted 14 degrees downward. The elbow joint picks up 4 degrees on Y; with it zeroed the wrist guide lands 1.6 cm off center, still inside the wrist. Accepted. Elbow pre-bend is 17.6 degrees, acceptable. Optional correction in 3.6.

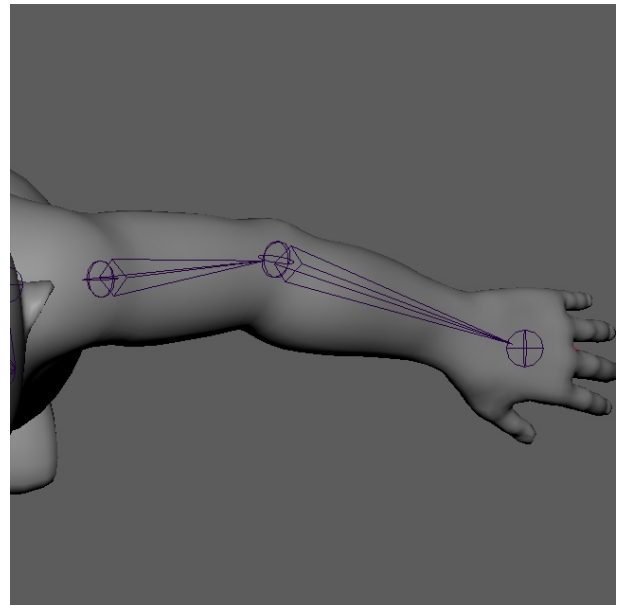
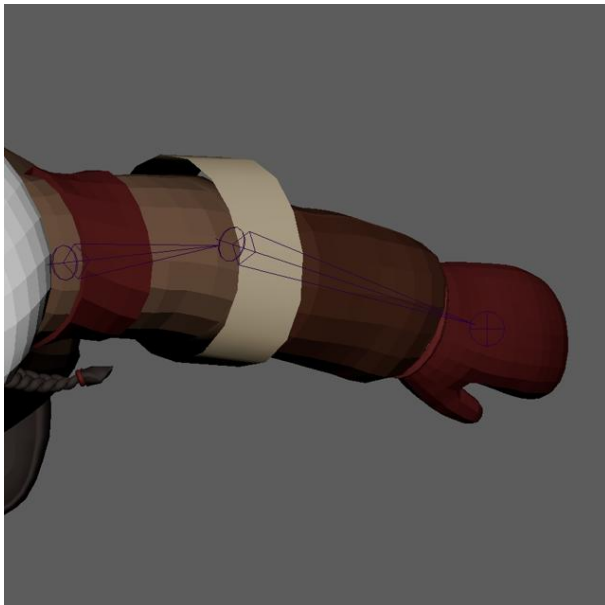
Both chains are symmetrical (left and right give the same numbers with mirrored signs). Positions used, in cm: hip 7.80, 60.91, -3.04; knee 8.22, 37.71, 1.84; ankle 7.69, 10.07, -3.32; shoulder 14.39, 96.87, -3.66; elbow 24.81, 85.14, -4.82; wrist 40.74, 69.63, 0.45.



Leg guides, side view. Knee ahead of the hip-ankle line (+Z).



Front view after zeroing the knee minor axes. Red locator: ankle landmark. Joint: where the chain lands.

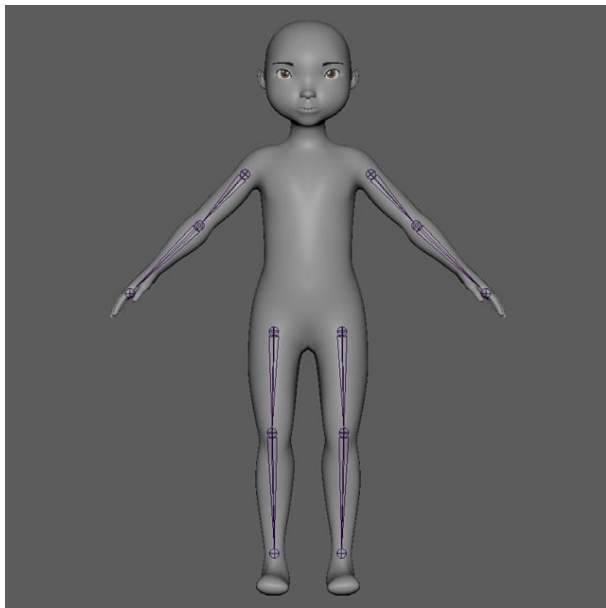


Arm guides, top view. Elbow behind the shoulder-wrist line (-Z).

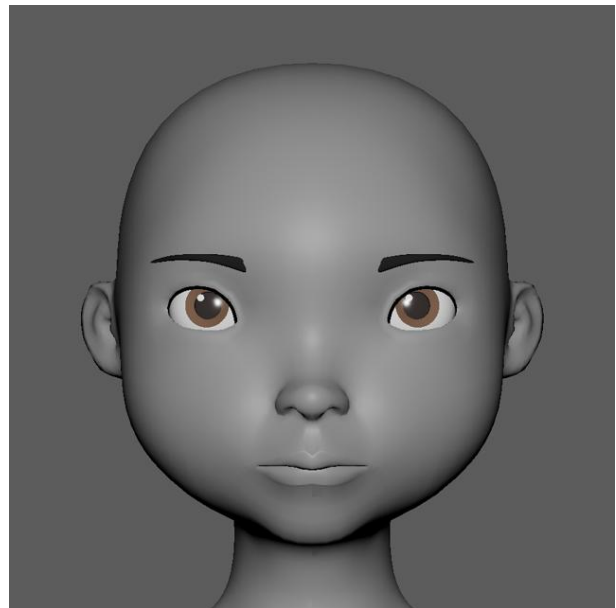
Top view after zeroing the elbow minor axes. Red locator: wrist landmark.

5. Checks passed

Check	Result
Body symmetry	5,870 vertices per side, 222 on the center line, maximum mirror deviation 0.000 cm.
Non-manifold, lamina faces	None on any mesh.
Construction history, intermediate shapes	None.
Vertex tweaks (pnts)	All zero.
Duplicate names, namespaces, unknown nodes, extra cameras, display layers, sets	None.
Unused shaders	None (37 shading groups, all assigned).
One shape per transform, no mesh parented under a mesh	OK. c_hair_GEO is a group (see naming).
UV sets	All meshes have a single set named map1.
Legs parallel, feet facing +Z	OK. Leg center X stays between 7.5 and 7.9 from hip to ankle. Knee direction: see section 4.
Neutral expression on lips, eyelids, eyebrows	OK.
Cloth and body elbows at the same place	OK visually. The sleeve cuff loop sits on the elbow line.
Triangles	Only on eye spheres (32 per sclera and pupil, 20 per specular) at the sphere poles. Acceptable, these meshes do not deform.



Body without clothes with the four guide chains.



Face, neutral expression.

6. Per-mesh summary

Mesh	Verts	Flags
c_body_GEO	11962	12 n-gons, 5 duplicate pairs, locked normals, pivot off origin
c_hood_GEO1	1155	2 n-gons, name, locked normals
c_pants_GEO	1530	2 n-gons, 1 duplicate pair, locked normals
c_button_GEO	794	2 duplicate vertices, locked normals
c_tunic_GEO	2227	locked normals
c_gloves_GEO	1388	locked normals
c_legwarmers_GEO	1860	locked normals
c_placeholder_bootBase	2400	unfrozen, name, locked normals

Mesh	Verts	Flags
c_bootCuffs_GEO	520	locked normals
c_ribbons_GEO	900	locked normals
c_furBalls_GEO	196	unfrozen, locked normals
c_armcuffs_GEO	504	locked normals
c_collar_GEO	794	locked normals
c_embroider_GEO	231	locked normals
c_circle_GEO	25	locked normals
c_bottomsSleeves_GEO	720	locked normals
c_furCuffs_GEO	108	locked normals
c_braids_GEO	4656	locked normals
c_bangs_GEO	1254	locked normals
c_hairties_GEO	800	locked normals
c_sidePieces_GEO	304	locked normals
c_ends_GEO	146	locked normals
L_sclera_GEO / R_sclera_GEO	1042	32 tris each, locked normals
L_pupil_GEO / R_pupil_GEO	898	32 tris, 3 materials each, locked normals
L_specular_GEO / R_specular_GEO	21	20 tris, R unfrozen, locked normals
c_eyelashes_GEO	108	unfrozen (t, r, s), locked normals
l_eyebrow_GEO / r_eyebrow_GEO	42	locked normals
c_topGums_GEO	1209	locked normals
c_lowGums_GEO	986	locked normals
C_tongue_GEO	235	name, locked normals
teeth (24 meshes)	114 to 182	locked normals, groups unfrozen

Resubmission: deliver Nini as a Maya ASCII scene with items 2.1 to 2.5 fixed. Rigging will re-run this check and skin a single jaw joint to validate the lip commissures before starting the rig.